

§6 可降阶的高阶微分方程

👉 $y^{(n)} = f(x)$ (I) 型的微分方程

👉 $y'' = f(x, y')$ (II) 型的微分方程

👉 $y'' = f(y, y')$ (III) 型的微分方程

定义 二阶及二阶以上的微分方程称为高阶微分方程

从本节起我们讨论高阶微分方程

有些高阶方程，可通过变量代换化成较低阶的方程求解

本节介绍三种容易降阶的高阶方程的求解方法.

一、 $y^{(n)} = f(x)$ (I) 型的微分方程

特点 方程右端 $f(x)$ 仅是 x 的函数.

解法 逐次积分

$$y^{(n)} = f(x)$$

$$y^{(n-1)} = \int f(x)dx + C_1$$

$$y^{(n-2)} = \int (\int f(x)dx)dx + C_1x + C_2$$

...

y 的表达式含有 n 个任意积分常数

例1 解方程 $y^{(3)} = \sin x + x$

解 逐次积分

$$y'' = -\cos x + \frac{x^2}{2} + C_1$$

$$y' = -\sin x + \frac{x^3}{2 \cdot 3} + C_1 x + C_2$$

$$y = -\cos x + \frac{x^4}{4!} + C_1 \frac{x^2}{2} + C_2 x + C_3$$

例2 求解初值问题:

$$\begin{cases} \frac{d^2 x}{dt^2} = \frac{F_0}{m} \left(1 - \frac{t}{T}\right) \\ x|_{t=0} = 0 \quad v = \frac{dx}{dt} \Big|_{t=0} = 0 \end{cases}$$

解 逐次积分

$$\frac{dx}{dt} = \frac{F_0}{m} \left(t - \frac{t^2}{2T}\right) + C_1 - (1)$$

$$x = \frac{F_0}{m} \left(\frac{t^2}{2} - \frac{t^3}{6T}\right) + C_1 t + C_2 - (2)$$

将初始条件:

$$x|_{t=0} = 0$$

$$v = \frac{dx}{dt} \Big|_{t=0} = 0 \text{ 代入}$$

$$\text{由(1)得: } C_1 = 0;$$

$$\text{由(2)得: } C_2 = 0$$

$$\begin{aligned} \therefore x(t) &= \frac{F_0}{m} \left(\frac{t^2}{2} - \frac{t^3}{6T}\right) \\ &= \frac{F_0 t^2}{2m} \left(1 - \frac{t}{3T}\right) \end{aligned}$$

二、 $y'' = f(x, y')$ (II)型的微分方程

特点 方程右端 $f(x, y')$ 不含 y

解法 令 $y' = P(x)$ 则 $y'' = \frac{dP}{dx} = P'$

方程(II)变为: $P' = f(x, P)$

关于变量
 x, P 的一阶
微分方程

应用前面一阶方程的求解, 设其通解:

$$P = \phi(x, C_1)$$

即, $\frac{dy}{dx} = \phi(x, C_1) \Rightarrow y = \int \phi(x, C_1) dx + C_2$

例3 求 : $(1+x^2)y'' = 2xy'$, 满足 $y|_{x=0} = 1, y'|_{x=0} = 3$
的特解

解 $\because$ 方程不显含 y

$\therefore$ 令 $y' = P, y'' = P'$

代入得 :

$$P' = \frac{2x}{1+x^2} P (P \neq 0)$$

积分得 :

$$\ln P = \ln(1+x^2) + \ln C_1$$

$$\Rightarrow P = (1+x^2)C_1$$

$$\text{即, } \frac{dy}{dx} = (1+x^2)C_1$$

再积分得:

$$y = C_1(x + \frac{x^3}{3}) + C_2$$

代初始条件得:

$$C_1 = 3, \quad C_2 = 1$$

$\therefore$ 所求特解为:

$$y = 3(x + \frac{x^3}{3}) + 1$$

例4 已知 $u(x, y, z) = u(\sqrt{x^2 + y^2 + z^2})$ 有连续的

二阶导数, 且满足 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$, 求 $u(x, y, z)$

解
$$\frac{\partial u}{\partial x} = u'(r) \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{xu'(r)}{r}$$

$$\frac{\partial u}{\partial y} = \frac{yu'(r)}{r} \quad \frac{\partial u}{\partial z} = \frac{zu'(r)}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u'(r)}{r} + x \frac{u''(r)r - u'}{r^2} \cdot \frac{x}{r} = \frac{u'}{r} + \frac{x^2}{r^3} (ru'' - u')$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3u'}{r} + \frac{ru'' - u'}{r^3} r^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3u'}{r} + \frac{ru'' - u'}{r^3} r^2 = 0$$

$$ru'' + 2u' = 0$$

$$u' = p \quad \frac{2}{r}p + p' = 0 \quad \frac{dp}{dr} = -\frac{2}{r}p \quad \frac{dp}{p} = -\frac{2}{r}dr$$

$$\ln p = -2\ln r + C_1 \quad p = \frac{C_1}{r^2}, \quad \frac{du}{dr} = \frac{C_1}{r^2} \quad u = -\frac{C_1}{r} + C_2$$

$$\text{即 } u(x, y, z) = \frac{C_1}{\sqrt{x^2 + y^2 + z^2}} + C_2$$

例5 设 $r = \sqrt{x^2 + y^2 + z^2} > 0$, 函数 $u = f(r)$ 具有二阶连续的导数, 且 $\operatorname{div}(\operatorname{grad} u) = \ln r$, 求 $f(r)$.

解

$$\operatorname{div}(\operatorname{grad} u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

$$\frac{\partial u}{\partial x} = f'(r) \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x f'(r)}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{f'(r)}{r} + x \frac{f'' r - f'}{r^2} \cdot \frac{x}{r} = \frac{f'}{r} + \frac{x^2}{r^3} (r f'' - f')$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3f'}{r} + \frac{r f'' - f'}{r^3} r^2 = \frac{2f' + r f''}{r}$$

以题意有 $2f' + r f'' = r \ln r$

$$\text{令 } f' = p,$$

$$p' + \frac{2}{r}p = \ln r \text{ 为一阶线性微分方程,}$$

$$p = e^{\int -\frac{2}{r} dr} (C_1 + \int \ln r \cdot e^{\int \frac{2}{r} dr} dr)$$

$$= \frac{1}{r^2} (C_1 + \int r^2 \ln r dr) = \frac{1}{r^2} (C_1 + \frac{1}{3} r^3 \ln r - \frac{1}{9} r^3)$$

$$f(r) = \int \frac{1}{r^2} (C_1 + \frac{1}{3} r^3 \ln r - \frac{1}{9} r^3) dr$$

$$\text{即 } f(r) = -\frac{C_1}{r} - \frac{1}{18} r^2 + \frac{1}{12} r^4 \ln r - \frac{1}{48} r^4 + C_2$$

Ex. 设 $f(u)$ 在 $(0, +\infty)$ 内具有二阶导数, 且 $z = f(\sqrt{x^2 + y^2})$

满足 $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, 若 $f(1)=0$, $f'(1)=1$, 求 $f(u)$ 的表达式.

解
$$\frac{\partial z}{\partial x} = f'(u) \frac{x}{\sqrt{x^2 + y^2}} = \frac{x f'(u)}{u}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{f'}{u} + \frac{x^2}{u^3} (u f'' - f')$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{f' + u f''}{u}$$

以题意有 $f' + u f'' = 0$ 解得通解为 $f(u) = C_1 \ln u + C_2$

代入初始条件得 $C_1 = 1, C_2 = 0$ 所以 $f(u) = \ln u$

例 6 求方程 $xy^{(5)} - y^{(4)} = 0$ 的通解.

解 设 $y^{(4)} = P(x)$, $y^{(5)} = P'(x)$

代入原方程 $xP' - P = 0$, ($P \neq 0$)

解线性方程, 得 $P = C_1 x$ 即 $y^{(4)} = C_1 x$,

两端积分, 得 $y''' = \frac{1}{2} C_1 x^2 + C_2, \dots \dots$,

$$y = \frac{C_1}{120} x^5 + \frac{C_2}{6} x^3 + \frac{C_3}{2} x^2 + C_4 x + C_5,$$

原方程通解为 $y = d_1 x^5 + d_2 x^3 + d_3 x^2 + d_4 x + d_5$

三、 $y'' = f(y, y')$ (III)型的微分方程

特点 方程右端 $f(y, y')$ 不含 x

解法 令 $y' = P(y)$ $y'' = \frac{dP}{dx} = \frac{dP}{dy} \frac{dy}{dx} = P \frac{dP}{dy}$

代入方程(III)得: $P \frac{dP}{dy} = f(y, P)$

关于变量
 y, P 的一
阶微分方
程。

设通解为 $P = y' = \phi(y, C_1)$

(III)的通解为: $\int \frac{dy}{\phi(y, C_1)} = x + C_2$

例7 求微分方程 $yy'' - y'^2 = 0$ 的通解

解 $\because$ 方程不显含 x

分离变量后积分:

$$\therefore \text{令 } y' = P, y'' = P \frac{dP}{dy}$$

$$\int \frac{dP}{P} = \int \frac{dy}{y} \text{ 得}$$

原方程变为:

$$P = C_1 y, \text{ 即 } y' = C_1 y,$$

$$yP \frac{dP}{dy} - P^2 = 0$$

再分离变量积分

$$\ln y = C_1 x + \ln C_2$$

若 $P \neq 0$, 方程成为:

$$\text{通解: } y = C_2 e^{C_1 x}$$

$$y \frac{dP}{dy} - P = 0$$

$P = 0$ 时, $y = C$ 时是原方程的解, 视 $y = C_2 e^{C_1 x}$ 之中 $C_1 = 0$.

另解 两端同乘不为零因子 $\frac{1}{y^2}$,

$$\frac{yy'' - y'^2}{y^2} = \frac{d}{dx} \left(\frac{y'}{y} \right) = 0, \quad \text{故 } y' = C_1 y,$$

从而通解为 $y = C_2 e^{C_1 x}$.

小结

类型

解法

$$y^{(n)} = f(x) - \text{(I)}$$

逐次积分

$$y'' = f(x, y') - \text{(II)}$$

$$\text{令 } y' = P$$

$$\Rightarrow \frac{dP}{dx} = f(x, P)$$

$$y'' = f(y, y') - \text{(III)}$$

$$\text{令 } y' = P$$

$$\Rightarrow P \frac{dP}{dy} = f(y, P)$$